

# Class Palindrome

java.lang.Object  
Palindrome

---

```
public class Palindrome  
extends Object
```

## Palindrome.java - Determine if a String is a palindrome.

**Problem Statement:** A palindrome is a word or phrase that reads the same forward and backward. Only letters and numbers are checked, spaces and punctuation are skipped. Case also is ignored.

Examples: Noon, Madam, 12321, and "Never odd or even!" are all palindromes.

Write a method with the heading:

```
public static boolean isPalindrome(String check) ;
```

which determines if a String entered by the user is a palindrome.

### Algorithm:

In **main**:

1. Prompt for a String to test. (If empty then quit.)
2. Call isPalindrome method using the String entered by the user as a parameter.
  - If isPalindrome returns "true" then tell the user the String is a palindrome.
  - Otherwise tell the user it isn't a palindrome.
3. Go back to step 1.

In the **isPalindrome** method:

Start a continuous loop:

1. If the length of the String is less than 2, then return "true".
2. Extract first and last characters from the String, converting both to uppercase.
3. Look at the first and last characters:
  - a. If the first character isn't a number or letter, then remove it from the String and restart the loop.
  - b. If the last character isn't a number or letter, then remove it from the String and restart the loop.
  - c. If the first and last characters are different, then return "false".
4. Remove the first and last characters from the String and restart the loop.

**Version:**

**Author:**

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## ***Constructor Summary***

### Constructors

| Constructor               | Description |
|---------------------------|-------------|
| <code>Palindrome()</code> |             |

## ***Method Summary***

### All Methods      Static Methods      Concrete Methods

| Modifier and Type      | Method                                       | Description |
|------------------------|----------------------------------------------|-------------|
| private static boolean | <code>isPalindrome(String testString)</code> |             |
| static void            | <code>main(String [] args)</code>            |             |

### Methods inherited from class `Object`

`clone` , `equals` , `finalize` , `getClass` , `hashCode` , `notify` , `notifyAll` , `toString` , `wait` , `wait` , `wait`

## ***Constructor Details***

### **Palindrome**

```
public Palindrome()
```

## ***Method Details***

### **main**

```
public static void main(String [] args)
```

### **isPalindrome**

```
private static boolean isPalindrome(String testString)
```